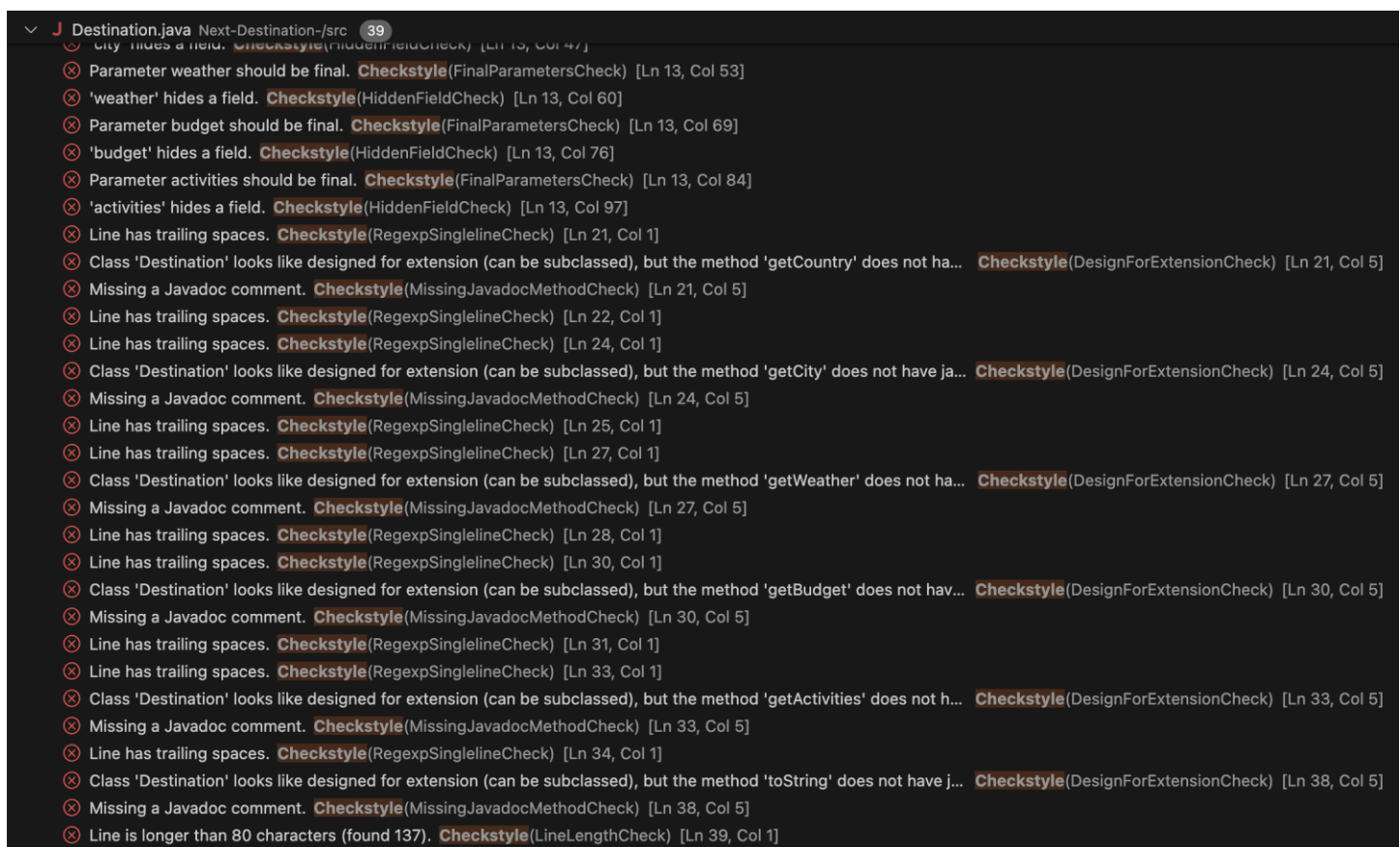


For this task, I used Checkstyle to analyze the code style and structure of my project. Checkstyle checks whether the code follows the rules defined in a style guide, in this case, the Google Java Style Guide. It helps spot problems like missing comments, bad formatting, or naming issues that could make the code hard to read or maintain.

Tools used:

1. Checkstyle for Java: This is a tool that enforces coding standards.
2. Checkstyle Extension in VS Code: I integrated the tool with VS Code to run checks directly in my editor.

Results:



The screenshot shows the Checkstyle extension in VS Code displaying a list of errors for the file `Destination.java`. The errors are listed in a scrollable panel on the right side of the editor. Each error is preceded by a red 'X' icon and includes the error message, the Checkstyle rule name, and the location (line and column) of the error.

```
✗ City hides a field. Checkstyle(HiddenFieldCheck) [Ln 13, Col 47]
✗ Parameter weather should be final. Checkstyle(FinalParametersCheck) [Ln 13, Col 53]
✗ 'weather' hides a field. Checkstyle(HiddenFieldCheck) [Ln 13, Col 60]
✗ Parameter budget should be final. Checkstyle(FinalParametersCheck) [Ln 13, Col 69]
✗ 'budget' hides a field. Checkstyle(HiddenFieldCheck) [Ln 13, Col 76]
✗ Parameter activities should be final. Checkstyle(FinalParametersCheck) [Ln 13, Col 84]
✗ 'activities' hides a field. Checkstyle(HiddenFieldCheck) [Ln 13, Col 97]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 21, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'getCountry' does not ha... Checkstyle(DesignForExtensionCheck) [Ln 21, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 21, Col 5]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 22, Col 1]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 24, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'getCity' does not have ja... Checkstyle(DesignForExtensionCheck) [Ln 24, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 24, Col 5]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 25, Col 1]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 27, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'getWeather' does not ha... Checkstyle(DesignForExtensionCheck) [Ln 27, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 27, Col 5]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 28, Col 1]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 30, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'getBudget' does not hav... Checkstyle(DesignForExtensionCheck) [Ln 30, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 30, Col 5]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 31, Col 1]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 33, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'getActivities' does not h... Checkstyle(DesignForExtensionCheck) [Ln 33, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 33, Col 5]
✗ Line has trailing spaces. Checkstyle(RegexpSinglelineCheck) [Ln 34, Col 1]
✗ Class 'Destination' looks like designed for extension (can be subclassed), but the method 'toString' does not have j... Checkstyle(DesignForExtensionCheck) [Ln 38, Col 5]
✗ Missing a Javadoc comment. Checkstyle(MissingJavadocMethodCheck) [Ln 38, Col 5]
✗ Line is longer than 80 characters (found 137). Checkstyle(LineLengthCheck) [Ln 39, Col 1]
```

Error	What it Means
'weather' hides a field	The parameter weather in a method has the same name as a class field. This can confuse the compiler and developers.
Missing Javadoc Comment	There's no Javadoc comment for the methods. Public methods should have comments explaining what they do.
Designed for Extension	Checkstyle warns that the class might be extended, but it doesn't explain how to do it safely.
Trailing Spaces	There are unnecessary spaces at the end of the line. These spaces don't do anything and just clutter the file.
Line Longer Than 80 Characters	The line exceeds 80 characters, which makes it hard to read.

What these errors mean:

1. Hidden Field Errors:

This happens when a method parameter has the same name as a class variable. It's confusing because it's not clear whether the code is referring to the parameter or the class variable.

2. Missing Javadoc Comments:

Public methods and classes are expected to have comments describing what they do. This helps others understand the code quickly.

3. Trailing Spaces:

These are extra spaces at the end of a line. They don't affect how the code runs but are considered bad style because they make the code less clean.

4. Line Length Violations:

Lines longer than 80 characters can be difficult to read. This is especially noticeable when reviewing code or using tools that wrap long lines.

5. Class Designed for Extension:

The tool warns that my class might be extended in the future, but there are no guidelines on how to extend it safely.

Checkstyle helped me find several code style problems, like missing comments, field-hiding issues, and formatting errors. These kinds of tools are useful for keeping code clean and readable.